

Machine Learning Quiz — Questions and Answers

Unified answer sheet with questions, correct choices, and option-by-option explanations.

1. Single-choice (6 Score)

1. Which of the following is a regression problem?

- A. Classifying emails as spam or not
- B. Predicting house prices
- C. Identifying handwritten digits
- D. Grouping similar articles

Correct answer: B

Option Explanations:

Option A: Incorrect. This is a classification task because the output is a discrete label: spam or not spam.

Option B: Correct. Regression predicts a continuous numerical value, and house prices are continuous.

Option C: Incorrect. Identifying digits is usually a multiclass classification problem.

Option D: Incorrect. Grouping similar articles is clustering, which is an unsupervised learning task.

2. Single-choice (7 Score)

2. What is the goal of clustering?

- A. Predict future values
- B. Separate data into labeled groups
- C. Group similar items
- D. Maximize model accuracy

Correct answer: C

Option Explanations:

Option A: Incorrect. Predicting future values is usually associated with forecasting or regression.

Option B: Incorrect. Clustering does not require pre-existing labels; it discovers structure automatically.

Option C: Correct. Clustering aims to place similar data points into the same group.

Option D: Incorrect. Clustering is not defined by predictive accuracy on labeled outcomes.

3. Single-choice (6 Score)

3. Clustering is an example of:

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Semi-supervised learning

Correct answer: B

Option Explanations:

Option A: Incorrect. Supervised learning requires labeled targets.

Option B: Correct. Clustering finds patterns or groups in unlabeled data, so it is unsupervised learning.

Option C: Incorrect. Reinforcement learning is based on actions, feedback, and rewards.

Option D: Incorrect. Semi-supervised learning uses a mix of labeled and unlabeled data, which is not the standard setting for clustering.

4. Single-choice (7 Score)

4. In K-Means, what does 'K' represent?

A. Number of features

B. Learning rate

C. Number of clusters

E. Number of layers

Correct answer: C

Option Explanations:

Option A: Incorrect. The number of features describes the dimension of each data point, not K.

Option B: Incorrect. Learning rate is used in optimization methods such as gradient descent.

Option C: Correct. In K-Means, K is the number of clusters to be formed.

Option E: Incorrect. Number of layers is a neural-network concept, not a K-Means parameter.

5. Single-choice (6 Score)

5. What is the objective of the K-Means algorithm?

A. Maximize inter-cluster similarity

B. Increase classification accuracy

C. Reduce overfitting

D. Minimize inner-cluster distance

Correct answer: D

Option Explanations:

Option A: Incorrect. K-Means tries to make points within the same cluster close, not maximize similarity between different clusters.

Option B: Incorrect. K-Means is not a classification algorithm.

Option C: Incorrect. Overfitting is mainly a supervised learning concern and is not the objective of K-Means.

Option D: Correct. K-Means minimizes within-cluster variance or, equivalently, the sum of squared distances from points to their assigned centers.

6. Single-choice (7 Score)

6. What is the role of an activation function in a neural network?

A. Store weights

B. Introduce non-linearity

C. Update gradients

D. Normalize inputs

Correct answer: B

Option Explanations:

Option A: Incorrect. Weights are parameters of the network, not stored by the activation function.

Option B: Correct. Activation functions introduce non-linearity, allowing neural networks to represent complex relationships.

Option C: Incorrect. Gradients are computed during backpropagation and then used by the optimizer.

Option D: Incorrect. Input normalization is a preprocessing or normalization-layer task, not the basic role of the activation function.

7. Single-choice (6 Score)

7. What does each layer in a neural network do?

A. Compresses data

B. Randomly splits inputs

C. Transforms inputs with weights and activation

D. Predicts final labels

Correct answer: C

Option Explanations:

Option A: Incorrect. Some architectures may reduce dimension, but compression is not the universal role of every layer.

Option B: Incorrect. Layers do not randomly split inputs as their main purpose.

Option C: Correct. A layer typically applies a weighted linear transformation followed by an activation function.

Option D: Incorrect. Usually only the output layer is directly responsible for final prediction.

8. Single-choice (7 Score)

8. Why is stochastic gradient descent used?

A. To speed up learning using mini-batches

B. To calculate exact gradients

C. To avoid local minima

D. To increase model size

Correct answer: A

Option Explanations:

Option A: Correct. SGD and mini-batch methods make learning scalable and efficient on large datasets.

Option B: Incorrect. SGD uses approximate gradients from samples or mini-batches, not exact full-dataset gradients.

Option C: Incorrect. Noise in SGD may sometimes help exploration, but avoiding local minima is not its primary purpose.

Option D: Incorrect. Optimization methods do not increase model size.

9. Single-choice (6 Score)

9. Which of the following about stochastic gradient descent (SGD) is correct?

- A. The larger the data batch is, the more effective SGD is
- B. The smaller the data batch is, the more effective SGD is
- C. SGD is used when the data size is very large
- D. SGD is less useful than the standard gradient descent method

Correct answer: C

Option Explanations:

- Option A:** Incorrect. Larger batches reduce gradient noise, but that alone does not make SGD universally more effective.
- Option B:** Incorrect. Smaller batches can speed iterations, but excessively small batches may make training unstable.
- Option C:** Correct. SGD is especially useful when the dataset is very large and full-batch gradient computation is too expensive.
- Option D:** Incorrect. SGD is often more practical than full-batch gradient descent in modern machine learning.

10. Single-choice (7 Score)

10. In backpropagation, what rule is used to compute gradients?

- A. Chain rule
- B. Product rule
- C. Sum rule
- D. L'Hopital's rule

Correct answer: A

Option Explanations:

- Option A:** Correct. Backpropagation efficiently applies the chain rule through the layers of the network.
- Option B:** Incorrect. The product rule is useful in calculus, but it is not the central principle behind backpropagation.
- Option C:** Incorrect. The sum rule alone is not what makes backpropagation work through deep compositions.
- Option D:** Incorrect. L'Hopital's rule is unrelated to standard neural-network gradient computation.

11. Single-choice (7 Score)

11. Which of the following is true about convex functions?

- A. They have multiple local minima
- B. Every local minimum is a global minimum
- C. They must be a quadratic function
- D. Their second derivative is always zero

Correct answer: B

Option Explanations:

Option A: Incorrect. A convex function does not have poor isolated local minima in the usual optimization sense.

Option B: Correct. For convex functions, any local minimum is also a global minimum.

Option C: Incorrect. Convex functions need not be quadratic; many other functions are convex as well.

Option D: Incorrect. A zero second derivative would indicate a linear function, not all convex functions.

12. Single-choice (7 Score)

12. A one-dimensional function $f(x): \mathbb{R} \rightarrow \mathbb{R}$ is convex if and only if:

- A. $f(x)$ is linear
- B. The function has a unique stationary point
- C. The line segment between any two points on the function curve always lies above the function
- D. The gradient is always decreasing

Correct answer: C

Option Explanations:

Option A: Incorrect. Linear functions are convex, but convexity is much broader than linearity.

Option B: Incorrect. A convex function need not have a unique stationary point in general.

Option C: Correct. Convexity means the chord between any two points on the graph lies above or on the graph.

Option D: Incorrect. For a differentiable convex function, the gradient is non-decreasing, not always decreasing.

13. Single-choice (7 Score)

13. What happens if the step size in gradient descent is too large?

- A. The algorithm converges faster
- B. The algorithm gets stuck in a local minimum
- C. The algorithm may overshoot and diverge
- D. The gradient becomes zero

Correct answer: C

Option Explanations:

Option A: Incorrect. A somewhat larger step can help, but if it is too large, convergence typically becomes unstable.

Option B: Incorrect. A large step size more often causes oscillation or divergence than getting stuck.

Option C: Correct. When the step size is too large, updates can overshoot the minimum and the algorithm may diverge.

Option D: Incorrect. The gradient is determined by the function landscape, not forced to zero by a large step size.

14. Single-choice (7 Score)

14. What is the typical convergence rate of gradient descent on a smooth and strongly convex function?

- A. Exponential e^t
- B. Sublinear $o(1/t)$
- C. Linear $O(1/t)$
- D. Quadratic $O(1/t^2)$

Correct answer: C

Option Explanations:

Option A: Incorrect. The method does not grow like e^t ; convergence means the error decays rather than explodes.

Option B: Incorrect. Smooth and strongly convex settings admit a faster rate than the usual sublinear convex rate.

Option C: Chosen answer. The intended idea is linear convergence. Strictly speaking, the standard textbook statement is geometric or linear convergence, often written as $O((1-\mu/L)^t)$, not $O(1/t)$.

Option D: Incorrect. Quadratic or $O(1/t^2)$ is not the typical rate for basic gradient descent in the strongly convex case.

15. Single-choice (7 Score)

15. Which of the following is true about the gradient at the minimum of a convex function?

- A. It is always negative
- B. It is equal to the Hessian
- C. It is equal to zero
- D. It is undefined

Correct answer: C

Option Explanations:

Option A: Incorrect. A gradient is a vector or derivative, not something that is simply 'always negative' at a minimum.

Option B: Incorrect. The gradient and Hessian are different objects: first derivative versus second derivative.

Option C: Correct. For a differentiable convex function, the gradient at an interior minimum is zero.

Option D: Incorrect. At a standard smooth minimum, the gradient is well-defined and equals zero.